

5(a)	line drawn from right atrium to the arrow into the pump and line drawn from the arrow out of the heater back to the (inferior / superior) vena cava ;	1
5(b)(i)	alveolar wall / alveolar epithelium ; A alveolus / alveoli A wall of air sacs	1
5(b)(ii)	any three from: oxygen moves (from the oxygen-enriched air) into the blood and carbon dioxide moves from the blood (into the oxygen-enriched air) / AW ; by diffusion / down their concentration gradients ; (opposite flow arrangement / counter current) maintains a steep diffusion gradient / ref. to equilibrium is not reached / AW ; short diffusion pathway (across the membrane) ; AVP ; e.g. oxygenator membrane (will have a) large surface area	3

Question	Answer	Marks
5(c)	(the tunica media of aorta) contains, more elastin / more elastic fibres / more elastic tissue / less smooth muscle, than the muscular artery ; ora <i>plus any two from:</i> allows the aorta to stretch / aorta less likely to burst (as a result of high blood pressure) ; during ventricular contraction / systole ; (after stretching) the aorta recoils ; maintains high blood pressure ; evens out the flow of blood / AW ;	3
5(d)(i)	for speeding up / as enzyme for / for catalysing, conversion of (water and) carbon dioxide to, carbonic acid / hydrogencarbonate ions ;	1
5(d)(ii)	hydrogencarbonate ions do not leave the red blood cells so no charge imbalance / AW ;	1